

Jobsheet 2

Data Warehouse



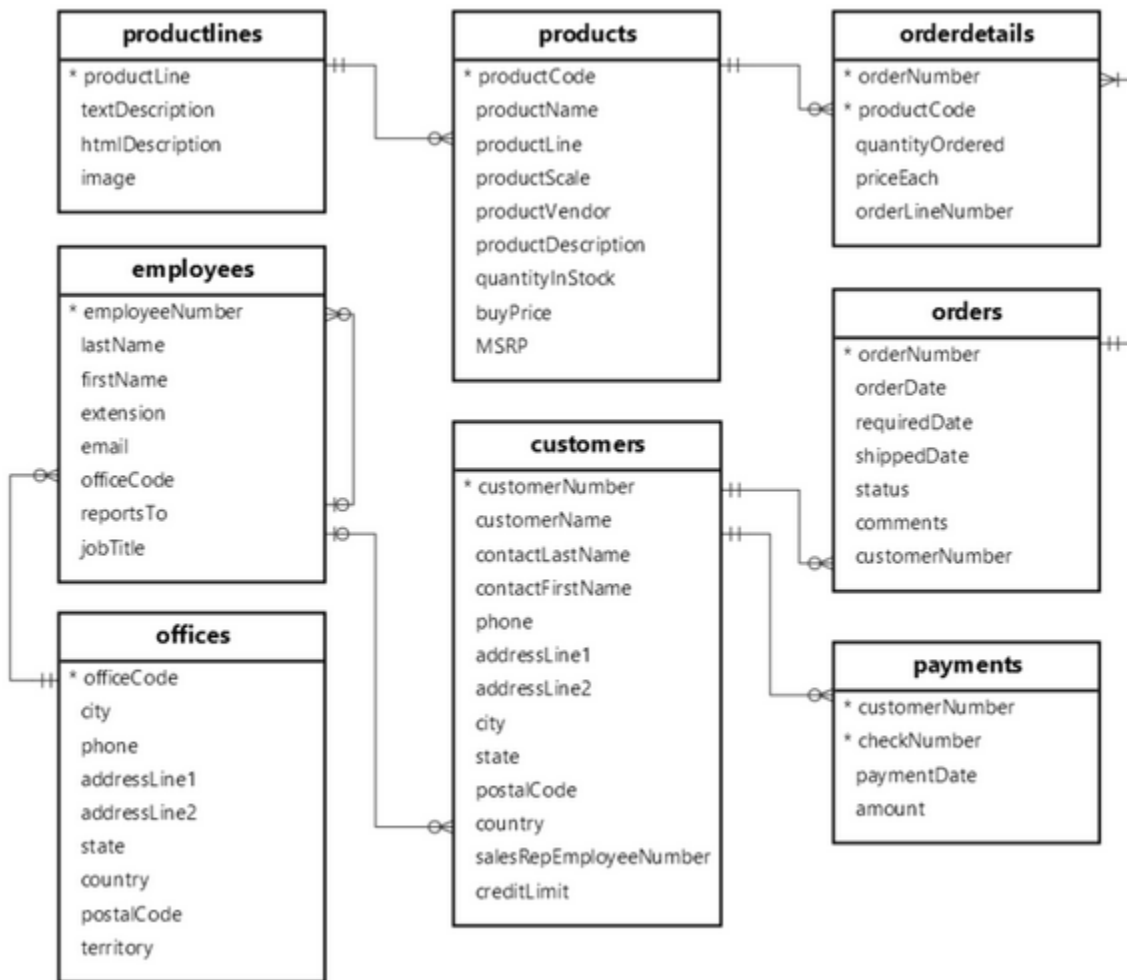
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STUDY PROGRAM D-IV BUSINESS INFORMATION SYSTEMS
DEPARTMENT OF INFORMATION TECHNOLOGY
STATE POLYTECHNIC OF MALANG
YEAR 2024

Jobsheet 2: Database Operasional

Studi Kasus

LegendVehicle merupakan perusahaan jual-beli tukar-tambah kendaraan klasik. Perusahaan ini memiliki cabang di berbagai negara. LegendVehicle memiliki sistem informasi ERP sendiri. Salah satu modul dari sistem ERP tersebut adalah modul penjualan. Desain database dari modul tersebut adalah sebagai berikut:



Selain itu proses penjualan kendaraan pada perusahaan tersebut bukan hanya melalui showroom cabang, melainkan reseller-reseller bebas lainnya.

Data penjualan dari cabang tersebut dapat diunduh melalui link berikut: [>> Data Penjualan <<](#)

Tugas 1

1. Import data perusahaan tersebut pada DBMS MySQL!

Table	Action	Rows	Type	Collation	Size	Overhead
☐ customers	★ Browse Structure Search Insert Empty Drop	122	InnoDB	utf8mb4_0900_ai_ci	64.0 KIB	-
☐ employees	★ Browse Structure Search Insert Empty Drop	23	InnoDB	utf8mb4_0900_ai_ci	48.0 KIB	-
☐ offices	★ Browse Structure Search Insert Empty Drop	7	InnoDB	utf8mb4_0900_ai_ci	16.0 KIB	-
☐ orderdetails	★ Browse Structure Search Insert Empty Drop	2,996	InnoDB	utf8mb4_0900_ai_ci	32.0 KIB	-
☐ orders	★ Browse Structure Search Insert Empty Drop	326	InnoDB	utf8mb4_0900_ai_ci	32.0 KIB	-
☐ payments	★ Browse Structure Search Insert Empty Drop	273	InnoDB	utf8mb4_0900_ai_ci	16.0 KIB	-
☐ productlines	★ Browse Structure Search Insert Empty Drop	7	InnoDB	utf8mb4_0900_ai_ci	16.0 KIB	-
☐ products	★ Browse Structure Search Insert Empty Drop	110	InnoDB	utf8mb4_0900_ai_ci	80.0 KIB	-
8 tables	Sum	3,864	InnoDB	utf8mb4_0900_ai_ci	304.0 KIB	0 B

2. Analisa struktur data dari database perusahaan tersebut, dalam bentuk tabel, analisa hubungan setiap tabel nya!

Tabel 1	Tabel 2	Jenis Relasi
Productlines	Products	One to many
Employees	Offices	One to many
Employees	Customers	One to many
Products	Orderdetails	One to many
Orderdetails	Orders	One to many
Orders	Customers	One to many
Customers	Payment	One to many

3. Analisa jumlah field pada setiap tabel!

Nama Tabel	Jumlah Field
Productlines	4
Products	9
Employees	8
Offices	9
Payments	4
Orderdetails	5
Orders	7
Customers	13

PRAKTIKUM 1

1. Jalankan **query** berikut pada **DBMS MySql** yang telah tersedia **data Perusahaan LegendVehicle**.

SELECT *

FROM employees employe, employees manager, customer cust

WHERE employee.reportsTo=manager.employeeNumber

AND employee.employeeNumber=cust.salesRepEmployeeNumber;

maka hasil dari query tersebut adalah data Employee beserta Manajernya dan Customer yang ia miliki. perhatikan hasil data dengan seksama.

employeeNumber	lastName	firstName	extension	email	officeCode	reportsTo	jobTitle	employeeNumber	lastName	firstName	extension	email	officeCode	reportsTo	jobTitle
1165	Jennings	Leslie	x3291	ljennings@classicmodelcars.com	1	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)
1165	Jennings	Leslie	x3291	ljennings@classicmodelcars.com	1	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)
1165	Jennings	Leslie	x3291	ljennings@classicmodelcars.com	1	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)
1165	Jennings	Leslie	x3291	ljennings@classicmodelcars.com	1	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)
1165	Jennings	Leslie	x3291	ljennings@classicmodelcars.com	1	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)
1165	Jennings	Leslie	x3291	ljennings@classicmodelcars.com	1	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)
1166	Thompson	Leslie	x4065	lthompson@classicmodelcars.com	1	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)
1166	Thompson	Leslie	x4065	lthompson@classicmodelcars.com	1	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)
1166	Thompson	Leslie	x4065	lthompson@classicmodelcars.com	1	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)
1166	Thompson	Leslie	x4065	lthompson@classicmodelcars.com	1	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)
1166	Thompson	Leslie	x4065	lthompson@classicmodelcars.com	1	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)
1166	Thompson	Leslie	x4065	lthompson@classicmodelcars.com	1	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)
1166	Thompson	Leslie	x4065	lthompson@classicmodelcars.com	1	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)

customerNumber	customerName	contactLastName	contactFirstName	phone	addressLine1	addressLine2	city	state	postalCode	country	salesRepEmployeeNumber	creditLimit
124	Mini Gifts Distributors Ltd.	Nelson	Susan	4155551450	5677 Strong St.	NULL	San Rafael	CA	97562	USA	1165	210500.00
129	Mini Wheels Co.	Murphy	Julie	6505555787	5557 North Pendale Street	NULL	San Francisco	CA	94217	USA	1165	64600.00
161	Technics Stores Inc.	Hashimoto	Juri	6505556809	9408 Furth Circle	NULL	Burlingame	CA	94217	USA	1165	84600.00
321	Corporate Gift Ideas Co.	Brown	Julie	6505551386	7734 Strong St.	NULL	San Francisco	CA	94217	USA	1165	105000.00
450	The Sharp Gifts Warehouse	Frick	Sue	4085553659	3086 Ingle Ln.	NULL	San Jose	CA	94217	USA	1165	77600.00
487	Signal Collectibles Ltd.	Taylor	Sue	4155554312	2793 Furth Circle	NULL	Brisbane	CA	94217	USA	1165	60300.00
112	Signal Gift Stores	King	Jean	7025551838	8489 Strong St.	NULL	Las Vegas	NV	83030	USA	1166	71800.00
205	Toys4GrownUps.com	Young	Julie	6265557265	78934 Hillside Dr.	NULL	Pasadena	CA	90003	USA	1166	90700.00
219	Boards & Toys Co.	Young	Mary	3105552373	4097 Douglas Av.	NULL	Glendale	CA	92561	USA	1166	11000.00
239	Collectable Mini Designs Co.	Thompson	Valarie	7605558146	361 Furth Circle	NULL	San Diego	CA	91217	USA	1166	105000.00
347	Men 'R' US Retailers, Ltd.	Chandler	Brian	2155554369	6047 Douglas Av.	NULL	Los Angeles	CA	91003	USA	1166	57700.00
475	West Coast Collectables Co.	Thompson	Steve	3105553722	3675 Furth Circle	NULL	Burbank	CA	94019	USA	1166	55400.00

employeeNumber	lastName	firstName	extension	email	officeCode	reportsTo	jobTitle	employeeNumber	lastName	firstName	extension	email	officeCode	reportsTo	jobTitle
1188	Firrelli	Julie	x2173	jfirrell@classicmodelcars.com	2	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Manager (NA)
1188	Firrelli	Julie	x2173	jfirrell@classicmodelcars.com	2	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)
1188	Firrelli	Julie	x2173	jfirrell@classicmodelcars.com	2	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)
1188	Firrelli	Julie	x2173	jfirrell@classicmodelcars.com	2	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)
1188	Firrelli	Julie	x2173	jfirrell@classicmodelcars.com	2	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)
1188	Firrelli	Julie	x2173	jfirrell@classicmodelcars.com	2	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)
1216	Patterson	Steve	x4334	spatterson@classicmodelcars.com	2	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)
1216	Patterson	Steve	x4334	spatterson@classicmodelcars.com	2	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)
1216	Patterson	Steve	x4334	spatterson@classicmodelcars.com	2	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)
1216	Patterson	Steve	x4334	spatterson@classicmodelcars.com	2	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)
1216	Patterson	Steve	x4334	spatterson@classicmodelcars.com	2	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)
1216	Patterson	Steve	x4334	spatterson@classicmodelcars.com	2	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)
1286	Tseng	Foon Yue	x2248	ftseng@classicmodelcars.com	3	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager

customerNumber	customerName	contactLastName	contactFirstName	phone	addressLine1	addressLine2	city	state	postalCode	country	salesRepEmployeeNumber	creditLimit
173	Collectables Co.	Tseng	Jerry	6175555555	4658 Baden Av.	NULL	Cambridge	MA	51247	USA	1188	43400.00
204	Online Mini Collectables	Barajas	Miguel	6175557555	7635 Spinnaker Dr.	NULL	Brickhaven	MA	58339	USA	1188	68700.00
320	Mini Creations Ltd.	Huang	Wing	5085559555	4575 Hillside Dr.	NULL	New Bedford	MA	50553	USA	1188	94500.00
339	Classic Gift Ideas, Inc.	Cervantes	Francisca	2155554695	782 First Street	NULL	Philadelphia	PA	71270	USA	1188	81100.00
379	Collectables For Less Inc.	Nelson	Allen	6175555855	7825 Douglas Av.	NULL	Brickhaven	MA	58339	USA	1188	70700.00
495	Diecast Collectables	Franco	Valarie	6175552555	6251 Ingle Ln.	NULL	Boston	MA	51003	USA	1188	85100.00
157	Diecast Classics Inc.	Leong	Kelvin	2155551555	7586 Pompton St.	NULL	Allentown	PA	70267	USA	1216	100600.00
198	Auto-Moto Classics Inc.	Taylor	Leslie	6175558428	16780 Pompton St.	NULL	Brickhaven	MA	58339	USA	1216	23000.00
286	Marta's Replicas Co.	Hernandez	Marta	6175558555	39323 Spinnaker Dr.	NULL	Cambridge	MA	51247	USA	1216	123700.00
362	Gifts4AllAges.com	Yoshido	Juri	6175559555	8616 Spinnaker Dr.	NULL	Boston	MA	51003	USA	1216	41900.00
363	Online Diecast Creations Co.	Young	Dorothy	6035558647	2304 Long Airport Avenue	NULL	Nashua	NH	62005	USA	1216	114200.00
462	FunGiftIdeas.com	Benitez	Violeta	5085552555	1785 First Street	NULL	New Bedford	MA	50553	USA	1216	85800.00
151	Muscle Machine Inc.	Young	Jeff	2125557413	4092 Furth Avenue	Suite 400	NYC	NY	10022	USA	1286	138500.00

2. Buka **tab baru** pada browser untuk melakukan eksekusi **query** berikut:

```
SELECT manager.employeeNumber as id_manager,
CONCAT(manager.firstName," ",manager.lastName) as Manager,
employee.employeeNumber as id_staff,
CONCAT(employee.firstName," ",employee.lastName) as staff
FROM employees employee, employees manager
WHERE employee.reportsTo=manager.employeeNumber
ORDER BY manager.firstName;
```

dari hasil **query** diatas maka akan ditemukan atasan dari setiap pegawai.

id_manager	Manager	id_staff	staff
1143	Anthony Bow	1165	Leslie Jennings
1143	Anthony Bow	1166	Leslie Thompson
1143	Anthony Bow	1188	Julie Firrelli
1143	Anthony Bow	1216	Steve Patterson
1143	Anthony Bow	1286	Foon Yue Tseng
1143	Anthony Bow	1323	George Vanauf
1002	Diane Murphy	1056	Mary Patterson
1002	Diane Murphy	1076	Jeff Firrelli
1102	Gerard Bondur	1337	Loui Bondur
1102	Gerard Bondur	1370	Gerard Hernandez
1102	Gerard Bondur	1401	Pamela Castillo
1102	Gerard Bondur	1501	Larry Bott
1102	Gerard Bondur	1504	Barry Jones
1102	Gerard Bondur	1702	Martin Gerard
1621	Mami Nishi	1625	Yoshimi Kato
1056	Mary Patterson	1088	William Patterson
1056	Mary Patterson	1102	Gerard Bondur
1056	Mary Patterson	1143	Anthony Bow
1056	Mary Patterson	1621	Mami Nishi
1088	William Patterson	1611	Andy Fixter
1088	William Patterson	1612	Peter Marsh
1088	William Patterson	1619	Tom King

Tugas 2

1. Gambarkan hirarki organisasi berdasarkan atasan dari setiap pegawai sesuai dengan hasil praktikum diatas!

```

SELECT
    manager.employeeNumber AS id_manager,
    CONCAT(manager.firstName, ' ', manager.lastName) AS manager_name,
    employee.employeeNumber AS id_staff,
    CONCAT(employee.firstName, ' ', employee.lastName) AS staff_name
FROM
    employees AS employee
JOIN
    employees AS manager
    ON employee.reportsTo = manager.employeeNumber
ORDER BY
    manager_name, staff_name;

```

id_manager	manager_name ▲ 1	id_staff	staff_name ▲ 2
1143	Anthony Bow	1286	Foon Yue Tseng
1143	Anthony Bow	1323	George Vanauf
1143	Anthony Bow	1188	Julie Firrelli
1143	Anthony Bow	1165	Leslie Jennings
1143	Anthony Bow	1166	Leslie Thompson
1143	Anthony Bow	1216	Steve Patterson
1002	Diane Murphy	1076	Jeff Firrelli
1002	Diane Murphy	1056	Mary Patterson
1102	Gerard Bondur	1504	Barry Jones
1102	Gerard Bondur	1370	Gerard Hernandez
1102	Gerard Bondur	1501	Larry Bott
1102	Gerard Bondur	1337	Loui Bondur
1102	Gerard Bondur	1702	Martin Gerard
1102	Gerard Bondur	1401	Pamela Castillo
1621	Mami Nishi	1625	Yoshimi Kato
1056	Mary Patterson	1143	Anthony Bow
1056	Mary Patterson	1102	Gerard Bondur
1056	Mary Patterson	1621	Mami Nishi
1056	Mary Patterson	1088	William Patterson
1088	William Patterson	1611	Andy Fixter
1088	William Patterson	1612	Peter Marsh
1088	William Patterson	1619	Tom King

2. Buka tab baru pada browser untuk melakukan eksekusi query berikut:
dari query tersebut menghasilkan jumlah customer dari setiap staff.

```
SELECT
    manager.employeeNumber AS id_manager,
    CONCAT(manager.firstName, " ", manager.lastName) AS Manager,
    employee.employeeNumber AS id_staff,
    CONCAT(employee.firstName, " ", employee.lastName) AS staff,
    COUNT(cust.customerNumber) AS total_cust
FROM
    employees AS employee
JOIN
    employees AS manager
    ON employee.reportsTo = manager.employeeNumber
LEFT JOIN
    customers AS cust
    ON employee.employeeNumber = cust.salesRepEmployeeNumber
GROUP BY
    employee.employeeNumber, manager.employeeNumber
ORDER BY
    manager.firstName;
```

id_manager	Manager	id_staff	staff	total_cust
1143	Anthony Bow	1165	Leslie Jennings	6
1143	Anthony Bow	1166	Leslie Thompson	6
1143	Anthony Bow	1188	Julie Firrelli	6
1143	Anthony Bow	1216	Steve Patterson	6
1143	Anthony Bow	1286	Foon Yue Tseng	7
1143	Anthony Bow	1323	George Vanauf	8
1002	Diane Murphy	1056	Mary Patterson	0
1002	Diane Murphy	1076	Jeff Firrelli	0
1102	Gerard Bondur	1337	Loui Bondur	6
1102	Gerard Bondur	1370	Gerard Hernandez	7
1102	Gerard Bondur	1401	Pamela Castillo	10
1102	Gerard Bondur	1501	Larry Bott	8
1102	Gerard Bondur	1504	Barry Jones	9
1102	Gerard Bondur	1702	Martin Gerard	6
1621	Mami Nishi	1625	Yoshimi Kato	0
1056	Mary Patterson	1088	William Patterson	0
1056	Mary Patterson	1102	Gerard Bondur	0
1056	Mary Patterson	1143	Anthony Bow	0
1056	Mary Patterson	1621	Mami Nishi	5
1088	William Patterson	1611	Andy Fixter	5
1088	William Patterson	1612	Peter Marsh	5
1088	William Patterson	1619	Tom King	0

Tugas 3

1. Siapakah staff dengan hirarki paling bawah yang berprestasi dilihat dari jumlah customer terbanyak?

id_staff	staff_name	total_cust
1401	Pamela Castillo	10

2. Jika KPI atasan dihitung dari customer yang dimilikinya dijumlah dengan customer dari staff dibawahnya, urutkan ranking prestasi keseluruhan pegawai beserta keterangan jumlah customer yang dimilikinya!

```
WITH RECURSIVE employee_hierarchy AS (  
  -- Level awal (staff tanpa bawahan)  
  SELECT  
    e.employeeNumber AS staff_id,  
    e.firstName AS staff_firstname,  
    e.lastName AS staff_lastname,  
    e.reportsTo AS manager_id,  
    COUNT(c.customerNumber) AS total_customers  
  FROM  
    employees e  
  LEFT JOIN  
    customers c ON e.employeeNumber = c.salesRepEmployeeNumber  
  GROUP BY  
    e.employeeNumber  
  
  UNION ALL  
  
  -- Rekursi untuk menghitung jumlah customer secara hierarki  
  SELECT  
    eh.manager_id AS staff_id,  
    m.firstName AS staff_firstname,  
    m.lastName AS staff_lastname,  
    m.reportsTo AS manager_id,  
    eh.total_customers  
  FROM  
    employee_hierarchy eh  
  JOIN  
    employees m ON eh.manager_id = m.employeeNumber  
)  
  
-- Menghitung total KPI (customer langsung + bawahan)  
SELECT  
  eh.staff_id,  
  CONCAT(eh.staff_firstname, ' ', eh.staff_lastname) AS staff_name,  
  SUM(eh.total_customers) AS kpi_total_customers  
FROM
```

```

employee_hierarchy eh
GROUP BY
  eh.staff_id, staff_name
ORDER BY
  kpi_total_customers DESC;

```

staff_id	staff_name	kpi_total_customers
1002	Diane Murphy	100
1056	Mary Patterson	100
1102	Gerard Bondur	46
1143	Anthony Bow	39
1088	William Patterson	10
1401	Pamela Castillo	10
1504	Barry Jones	9
1323	George Vanauf	8
1501	Larry Bott	8
1286	Foon Yue Tseng	7
1370	Gerard Hernandez	7
1165	Leslie Jennings	6
1166	Leslie Thompson	6
1188	Julie Firrelli	6
1216	Steve Patterson	6
1337	Loui Bondur	6
1702	Martin Gerard	6
1611	Andy Fixter	5
1612	Peter Marsh	5
1621	Mami Nishi	5
1076	Jeff Firrelli	0
1619	Tom King	0
1625	Yoshimi Kato	0

3. Analisa kembali data LegendVehicle untuk mendapatkan ranking pegawai berdasarkan KPI "Jumlah omset yang didapat". Urutkan ranking pegawai beserta keterangan dana yang didapat!

```

SELECT
  e.employeeNumber AS staff_id,
  CONCAT(e.firstName, ' ', e.lastName) AS staff_name,
  SUM(od.quantityOrdered * od.priceEach) AS total_revenue
FROM
  employees e
JOIN
  customers c ON e.employeeNumber = c.salesRepEmployeeNumber
JOIN
  orders o ON c.customerNumber = o.customerNumber
JOIN
  orderdetails od ON o.orderNumber = od.orderNumber
GROUP BY
  e.employeeNumber, staff_name

```

ORDER BY total_revenue DESC;		
staff_id	staff_name	total_revenue ▼ 1
1370	Gerard Hernandez	1258577.81
1165	Leslie Jennings	1081530.54
1401	Pamela Castillo	868220.55
1501	Larry Bott	732096.79
1504	Barry Jones	704853.91
1323	George Vanauf	669377.05
1612	Peter Marsh	584593.76
1337	Loui Bondur	569485.75
1611	Andy Fixter	562582.59
1216	Steve Patterson	505875.42
1286	Foon Yue Tseng	488212.67
1621	Mami Nishi	457110.07
1702	Martin Gerard	387477.47
1188	Julie Firrelli	386663.20
1166	Leslie Thompson	347533.03

4. Jika KPI yang pertama merupakan "Jumlah customer yang bertransaksi" sedangkan KPI yang kedua "Jumlah omset yang didapat". Maka, berapakah jumlah field yang dibutuhkan untuk mendapatkan informasi tersebut?

KPI	Number of Fields
Number of customers who transacted	3 fields (employeeNumber, EmployeeName, total_customer)
Total revenue obtained	4 fields (employeeNumber, employee_name, total_orders (handled by employee), total_revenue)

5. Buatlah report pertahun untuk KPI "Jumlah omset yang didapat" pada Foon Yue Tseng dan Pamela Castillo. Serta gambarkan grafiknya (grafik garis).

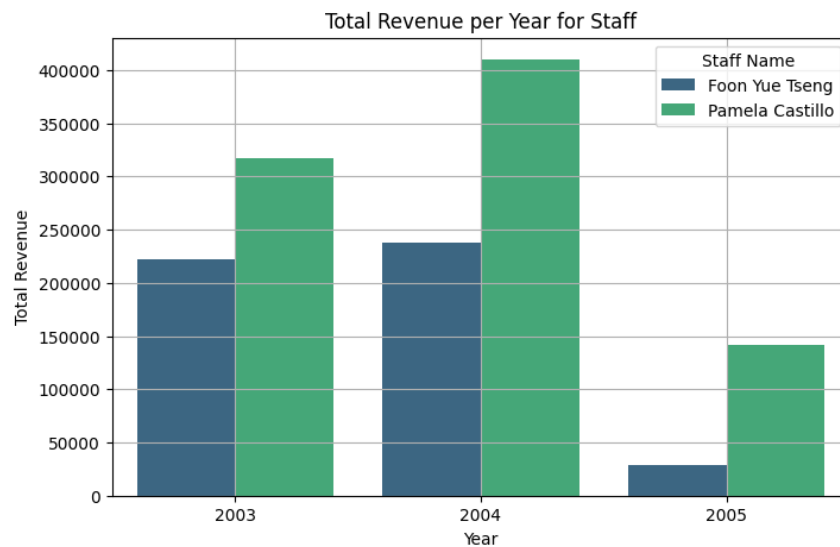
SELECT CONCAT(e.firstName, ' ', e.lastName) AS staff_name, YEAR(o.orderDate) AS year, SUM(od.quantityOrdered * od.priceEach) AS total_revenue FROM employees e

```

JOIN
  customers c ON e.employeeNumber = c.salesRepEmployeeNumber
JOIN
  orders o ON c.customerNumber = o.customerNumber
JOIN
  orderdetails od ON o.orderNumber = od.orderNumber
WHERE
  e.firstName = 'Foon Yue' OR e.firstName = 'Pamela'
GROUP BY
  staff_name, year
ORDER BY
  staff_name, year;

```

staff_name ▲ 1	year ▲ 2	total_revenue
Foon Yue Tseng	2003	221887.03
Foon Yue Tseng	2004	237255.26
Foon Yue Tseng	2005	29070.38
Pamela Castillo	2003	317104.78
Pamela Castillo	2004	409910.07
Pamela Castillo	2005	141205.70



Studi Kasus

Pak Huhut merupakan pemegang saham LegendVehicle. dia membutuhkan dashboard untuk melihat perkembangan penjualan (omset) disetiap cabang di tiap tahunnya. Dikarenakan perusahaan tersebut belum merekrut Data Engineer maka, penarikan informasi hanya bisa dilakukan melaluai OLTP yang ada.

Analisalah terlebih dahulu:

1. Field apa saja yang diperlukan untuk menampilkan penjualan di setiap cabang.

cabang → Nama cabang perusahaan.

tahun_penjualan → Tahun dari tanggal transaksi.

total_omset → Total penjualan (omset) yang dihitung dari $\text{quantityOrdered} \times \text{priceEach}$.

2. Bentuk query dengan memperhatikan relasi antar tabel.

```
SELECT
  o.city AS cabang,
  YEAR(od.orderDate) AS tahun_penjualan,
  SUM(odt.quantityOrdered * odt.priceEach) AS total_omset
FROM
  orders od
JOIN
  orderdetails odt ON od.orderNumber = odt.orderNumber
JOIN
  customers c ON od.customerNumber = c.customerNumber
JOIN
  employees e ON c.salesRepEmployeeNumber = e.employeeNumber
JOIN
  offices o ON e.officeCode = o.officeCode
GROUP BY
  o.city, YEAR(od.orderDate)
ORDER BY
  tahun_penjualan, total_omset DESC
LIMIT 0, 25;
```

cabang	tahun_penjualan ▲ 1	total_omset ▼ 2
Paris	2003	969959.90
London	2003	549551.94
San Francisco	2003	532681.13
NYC	2003	391175.53
Sydney	2003	304949.11
Boston	2003	301781.38
Tokyo	2003	267249.40
Paris	2004	1465229.84
London	2004	706014.52
NYC	2004	665317.99
Sydney	2004	542996.02
San Francisco	2004	517408.62
Boston	2004	467177.07
Tokyo	2004	151761.45
Paris	2005	648571.84
San Francisco	2005	378973.82
Sydney	2005	299231.22
London	2005	181384.24
Boston	2005	123580.17
NYC	2005	101096.20
Tokyo	2005	38099.22

SOAL BONUS: buatlah report lain dengan sumber data OLTP yang sama, analisa field yang digunakan, bentuk struktur query dan tuliskan dalam tabel serta grafiknya.

Required Fields:

- productLine (Product category) – Table products
- orderDate (Transaction date) – Table orders
- quantityOrdered and priceEach (Turnover calculation) – Table orderdetails
- productCode and orderNumber (For table relations)

```

SELECT
  p.productLine AS kategori_produk,
  YEAR(o.orderDate) AS tahun_penjualan,
  SUM(od.quantityOrdered * od.priceEach) AS total_omset
FROM
  orders o
JOIN
  orderdetails od ON o.orderNumber = od.orderNumber
JOIN
  products p ON od.productCode = p.productCode
GROUP BY

```

p.productLine, YEAR(o.orderDate)
ORDER BY
tahun_penjualan, total_omset DESC;

kategori_produk	tahun_penjualan ▲ 1	total_omset ▼ 2
Classic Cars	2003	1374832.22
Vintage Cars	2003	619161.48
Trucks and Buses	2003	376657.12
Motorcycles	2003	348909.24
Planes	2003	309784.20
Ships	2003	222182.08
Trains	2003	65822.05
Classic Cars	2004	1763136.73
Vintage Cars	2004	854551.85
Motorcycles	2004	527243.84
Planes	2004	471971.46
Trucks and Buses	2004	465390.00
Ships	2004	337326.10
Trains	2004	96285.53
Classic Cars	2005	715953.54
Vintage Cars	2005	323846.30
Motorcycles	2005	245273.04
Trucks and Buses	2005	182066.45
Planes	2005	172881.88
Ships	2005	104490.16
Trains	2005	26425.34

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Data sesuai tabel
data = {
    'kategori_produk': ['Classic Cars', 'Vintage Cars', 'Trucks and Buses', 'Motorcycles', 'Planes', 'Ships', 'Trains'] * 3,
    'tahun_penjualan': [2003]*7 + [2004]*7 + [2005]*7,
    'total_omset': [
        1374832.22, 619161.48, 376657.12, 348909.24, 309784.20, 222182.08, 65822.05,
        1763136.73, 854551.85, 465390.00, 527243.84, 471971.46, 337326.10, 96285.53,
        715953.54, 323846.30, 182066.45, 245273.04, 172881.88, 104490.16, 26425.34
    ]
}

# Konversi ke DataFrame
df = pd.DataFrame(data)

# Visualisasi
plt.figure(figsize=(12, 6))
sns.barplot(x='tahun_penjualan', y='total_omset', hue='kategori_produk', data=df)
plt.title("Perkembangan Penjualan (Omset) per Kategori Produk")
plt.xlabel("Tahun Penjualan")
plt.ylabel("Total Omset")
plt.legend(title="Kategori Produk", bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
plt.show()
```

